

WiShield: Privacy against Wi-Fi Human Tracking

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Abstract—Wi-Fi signals contain information about the surrounding propagation environment and have been widely used in various sensing applications such as gesture recognition, respiratory monitoring, and indoor position. Nevertheless, this information can also be easily stolen by eavesdroppers to obtain private information. In this paper, we propose WiShield, a new framework that protects legitimate users using Wi-Fi sensing applications while preventing unauthorized privacy attacks. The implementation of WiShield is based on a simple principle of physically encrypting Wi-Fi channel status information (CSI) to prevent eavesdroppers from inferring sensitive information through stolen CSI. To achieve a balance between encryption strength, sensing accuracy, and communication quality, we design an efficient multi-objective optimization framework that can safely deliver decryption keys to legitimate users and prevent illegal eavesdropping by eavesdroppers. We implemented the WiShield prototype on an SDR platform and conducted extensive experiments to verify its effectiveness in common Wi-Fi sensing applications. We believe that the implementation of WiShield can improve the privacy standards of Wi-Fi sensing applications, and it is also an important step towards making the integration of Integrated Sensing and Communications (ISAC).

Index Terms—Mobile and wireless security, Wi-Fi sensing, privacy protection.

I. INTRODUCTION

IN recent years, with the increasingly widespread deployment of Wi-Fi networks, Wi-Fi-based positioning [1]–[8] and sensing technologies [9]–[17] have developed rapidly. Unfortunately, while Wi-Fi devices bring us convenience,

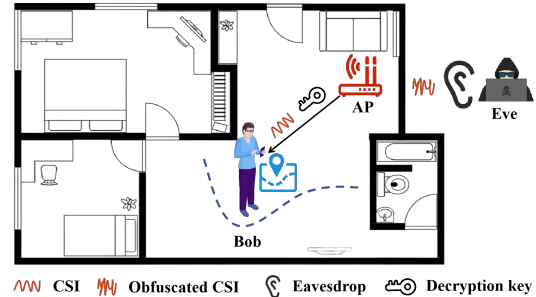


Fig. 1: Vision of WiShield: Preserving the legitimate user's (Bob) communication and sensing capabilities while preventing privacy attacks from the eavesdropper (Eve) by securely delivering the decryption key.

these ubiquitous radio frequency (RF) signals also bring us real security and privacy risks. We are drowned in these RF signals (in particularly channel state information (CSI) [18]), which can easily be eavesdropped on by hackers on our location, movement, and other private physiological information (such as breathing and heartbeat). Thieves could utilize this sensing capability to identify vacant homes or mine mobility and activity tracking data on a large scale to gain insights into the patterns of victim behavior. Such privacy breaches could easily lead to abuses such as targeted advertising or health insurance pricing.

Although this privacy threat has been recognized for many years [19]–[21], existing defense strategies [22]–[24] are difficult to resist sensing attacks while ensuring communication quality. PhyCloak [22] proposes a signal obfuscation technology to distort the private physical information leaked in the communication signal to protect the sensing power of only a legitimate user. IRShield [24] utilizes Intelligent Reflecting Surfaces (IRSs) positioned alongside an Access Point (AP) to disrupt the CSI, thereby rendering all sensing attempts ineffective. Unfortunately, this proposal contradicts the prevailing direction of evolving Wi-Fi towards ISAC (Integrated Sensing and Communications) [25], which emphasizes accommodating legitimate sensing users. Another option is to retrofit homes into Faraday cages by covering their walls with metal or tin foil. This solution effectively inhibits the transmission of wireless signals between adjacent residences but also restricts the operation of other necessary signals, such as cellular communication, within the household premises. As a result, existing solutions either defend against sensing attacks at the expense of communication quality or fail to support multiple legitimate users.

In this paper, we present WiShield, a novel system for

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defending against Wi-Fi sensing attacks while simultaneously supporting multiple legitimate users. As illustrated in Fig. 1, we use the Multiple-Input Multiple-Output (MIMO) capability of Wi-Fi to physically encrypt the CSI received by the device of the sensing subject. We will pass the key to legitimate sensing users to protect their use of Wi-Fi CSI's ability to perform sensing applications. On one hand, WiShield is fully compatible with existing Wi-Fi standards, so no additional hardware costs are required. On the other hand, the Wi-Fi communication quality of legitimate users can be well preserved.

WiShield has three key advantages compared to existing works. First, WiShield effectively limits the attacker's inference of Wi-Fi sensing data. By obfuscating Wi-Fi CSI, our approach will lead to incorrect results in inferences such as location, number of people, physiological status of occupants, and health indicators. Second, since WiShield itself does not transmit any signal, our defense is difficult to detect, and it is challenging for an attacker to break this defense without obtaining the key. Finally, we have made WiShield perfectly compatible with the current Wi-Fi framework in the form of software, and WiShield can launch defense while protecting the communication quality of legitimate users.

Nonetheless, two major challenges still exist in the implementation of WiShield. The first challenge is how to design a channel encryption and decryption algorithm that is compatible with existing Wi-Fi systems to meet actual multi-user needs. For defense against sensing attacks, existing security solutions often resort to arbitrary channel scrambling. In fact, there exists a distinct difference between regular communication security and sensing security: the former deals with digital plaintexts, while the latter involves physical plaintexts. Therefore, we can formally encrypt them in a physical manner to prevent Eve's eavesdropping without causing harm to Bob. Specifically, we can leverage (digital) Wi-Fi MIMO beamformers [26] to physically alter the transmission characteristics of the physical plaintext LTS. As long as the sequence of beamformers (equivalent to an encryption key) remains confidential, our physical plaintexts are effectively encrypted and thus concealed from Eve, who lacks knowledge of the key.

The second challenge is how to extract various parameters from CSI after decryption to support sensing applications. To solve this challenge, we use maximum likelihood estimation (MLE) to estimate parameters from the decrypted CSI. It is worth noting that WiShield does not need to know the specific location of the eavesdropper from beginning to end. We think WiShield has done an excellent job raising the bar for privacy in passive tracking systems. We recommend integrating the WiShield system into future hardware systems to defend against the threat of perceived attacks while protecting the quality of communication for legitimate users. In summary, the key contributions of WiShield include:

- We have implemented a Wi-Fi sensing attack defense system that can be used in multi-user scenarios. WiShield can prevent unauthorized privacy attacks while ensuring the communication capabilities of legitimate users.

- We design a source-defined channel encryption scheme to uphold compatibility with Wi-Fi, all while offering adaptable control to cater to the requirements of all participants in multi-user scenarios.
- We used the MLE algorithm to perform parameter estimation from the decrypted CSI and used these parameters for Wi-Fi based sensing applications.
- We implemented the WiShield prototype on USRP and conducted extensive experiments. The experimental results proved that WiShield can resist sensing attacks without affecting the communication quality of legitimate users.

The remainder of this paper is organized as follows. We first briefly touch upon related works in Section II. In Section III, we provide an overview of the background and outline the threat model that forms the basis of our work. Section IV delves into the intricate details of WiShield's system design. Following that, Section V elaborates on the implementation of WiShield, while Section VI presents a comprehensive analysis of the evaluations conducted on the system. In Section VII, we provide discussions of WiShield system. Finally, we draw our paper to a close in Section VIII with our concluding remarks.

II. RELATED WORKS

In this section, we first introduce related work based on privacy security caused by wireless sensing technology. We then summarize existing strategies to combat these sensing attacks. Finally, we outline our advantages and differences compared with previous strategies.

Wi-Fi sensing has attracted widespread attention due to its widely deployed infrastructure equipment. However, this convenience also brings serious security risks. The most common sensing attack is keystroke eavesdropping on mobile devices (mobile phones, tablets, etc.). WiKey [27] first used the changes in CSI due to keystrokes to perform keystroke inference. Then WindTalker [19] extended this keystroke reasoning to digital passwords on mobile devices and, combined with the use of deep learning, achieved more accurate keystroke reasoning. Next, Fang et al. [28] began to seek innovation in inference algorithms. They used English language structure analysis as the starting point to conduct text semantic inference based on CSI. Recently, WiPOS [29] considered the risk of key eavesdropping on fixed-position POS terminals exposed to Wi-Fi environments in shopping malls. SpiderMon [30] provides a new scenario and begins to try to use signals transmitted by commercial cell towers for passive ATM keystroke password eavesdropping. WINK [31] designed a training-free password reasoning algorithm by taking advantage of the spatiotemporal characteristics of the keystroke process. With the promotion of new Wi-Fi standards, WiKI-Eve [20] began to use Wi-Fi beamforming feedback information (BFI). Compared with CSI, which requires hardware cracking to obtain, the characteristics transmitted by BFI [32] clear-text have more serious security risks.

Another type of sensing attack focuses on human trajectory and location. By stealing Wi-Fi physical layer information, human movement trajectories can be inferred from channel

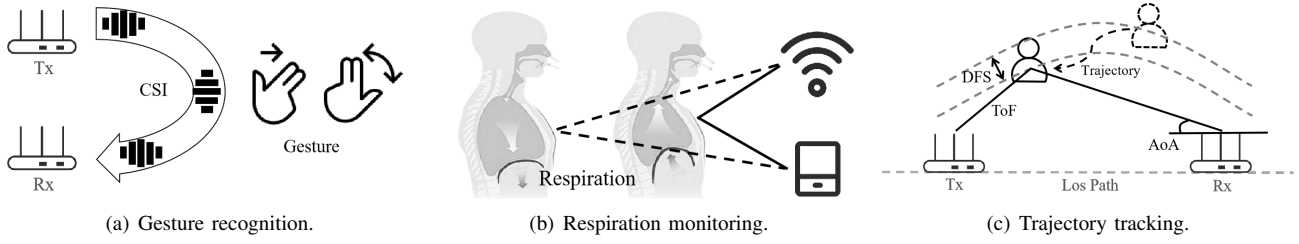


Fig. 2: Three typical Wi-Fi sensing applications. Wi-Fi sensing applications mainly utilize CSI changes in the channel environment for human activity tracking.

information. Banerjee et al. [33] implemented sensing attacks on MIMO-capable Wi-Fi routers using sliding windows as a metric of channel changes, focusing on detecting LOS crossovers. Zhu et al. [23] used a smartphone to track the entire indoor motion trajectory and even identify the motion of the target in a specific part. In addition, Camurati et al. [34] used a large number of experiments to prove that long-range hardware-level side-channel attacks can be achieved using radio frequency signals from Wi-Fi chipsets.

To combat such ubiquitous sensing attacks, existing defense solutions are mainly based on channel confusion. Qiao et al. [22] use full-duplex radio to inject random signals into the RF signal to resist sensing attacks. However, full-duplex implementation requires high hardware costs and complex system architecture to offset self-interference. Yao et al. [35] also proposed and extended this signal obfuscation strategy and used several moving parts (rotating antennas, fast fans, etc.) to optimize the hardware structure. IRShield [24] physically disrupts CSI by installing intelligent reflective surfaces (RIS) near the AP and injects false data traffic into the channel information to resist sensing attacks. But this kind of indiscriminate physical defense can interfere with legitimate users' sensing applications. 802.11 az [36] uses modern encryption and authentication standards such as WPA3 (Wi-Fi Protected Access 3) to protect network communications. Compared with past standards, WPA3 provides more robust encryption algorithms and more secure authentication methods to help prevent unauthorized access and data leakage. However, this privacy protection strategy has not been applied to LTS, which is still widely used.

To solve these problems, we propose WiShield, which uses MIMO technology [37] of Wi-Fi hardware to disrupt the channel, preventing ubiquitous sensing attacks while retaining the ability of legitimate users to sense and communicate. Of course, the implementation of WiShield requires hardware or firmware reconstruction, which also requires a certain hardware cost. Fortunately, implementing the ISAC framework by modifying the Wi-Fi architecture [38] is increasingly considered an ideal development path.

III. BACKGROUND AND ATTACK MODEL

In this section, we provide the basic principles of Wi-Fi sensing and several commonly used CSI-based sensing applications. Then, we define attack models to characterize the potential risks of Wi-Fi sensing to privacy. Finally, we briefly

introduce the rationale of our defense strategy and feasibility study.

A. Wi-Fi Tracking Basics

We first discuss the basic principles of Wi-Fi sensing. Wi-Fi sensing is a technology that uses the channel state information of Wi-Fi signals to implement various sensing applications. Specifically, the presence of moving targets in the environment will significantly change the signal propagation environment. Essentially, the basic principle of Wi-Fi sensing lies in the correlation between the change of Wi-Fi channel and the state of motion of objects.

Existing Wi-Fi sensing applications are mainly based on Long training sequence (LTS) [39] signal. LTS is an important concept in the Wi-Fi (IEEE 802.11) standard and is used to help Wi-Fi systems in After receiving the signal, the receiving end performs synchronization and channel state estimation. Unfortunately, Wi-Fi LTS is not encrypted and cannot be changed, so anyone can eavesdrop on LTS using a device equipped with a Wi-Fi card without being registered to the Wi-Fi network. When a Wi-Fi transmitter (Tx) sends a LTS signal $s_{n,m}$ (in frequency domain) at the n -th OFDM subcarrier (cardinality N) of the m -th packet (cardinality M), the receiving (Rx) signals become $y_{n,m} = h_{n,m}s_{n,m}$ where the $h_{n,m}$ represents the CSI of a Wi-Fi channel. To obtain the CSI, one often performs zero-forcing channel estimation algorithm [40] to deduce $y_{n,m} = h_{n,m}/s_{n,m}$. According to [41], the CSI $h_{n,m}$ can be modeled as:

$$h_{n,m} = \sum_{k=1}^K \alpha_{n,m,k} e^{-j2\pi(f_c + n\Delta f)\tau_{m,k}}, \quad (1)$$

where $\alpha_{n,m,k}$ is the attenuation along the k -th path, f_c denotes the carrier frequency, Δf represents the subcarrier spacing, and $\tau_{m,k} = \tau_k^S + \tau_{m,k}^D$ refers to the time delay: the static one τ_k^S from an arbitrary reflector and the dynamic one $\tau_{m,k}^D$ caused by its motions. CSI provides a large amount of information about the state of the nearby environment affected by the subject and is therefore widely used in indoor sensing applications [4]. For example, to realize indoor sensing applications, there are mainly three types of applications: gesture recognition, respiration monitoring, and trajectory tracking. The first common Wi-Fi application is gesture recognition, which infers and classifies the corresponding actions by analyzing the Wi-Fi channel changes caused by different

gestures (see in Fig. 2(a)). As illustrated in Fig. 2(b), the Wi-Fi-based physical sign monitoring system uses the changes in the chest cavity during breathing in the channel environment and combines it with advanced signal processing algorithms for respiration monitoring. Another type of application uses CSI parameters, such as angle of arrival (AoA), time of flight (ToF), and Doppler frequency shift (DFS) for trajectory tracking (see in Fig. 2(c)).

B. Threat Model

Although Wi-Fi-based indoor tracking is promising due to the widespread deployment of Wi-Fi devices, it also brings privacy concerns. As mentioned in Section III-A, existing Wi-Fi standards mainly use LTS to perform CSI estimation and communication symbol decoding. Unfortunately, Wi-Fi LTS is information without encryption [42], So anyone in the same Wi-Fi environment can use a device equipped with a Wi-Fi network card to eavesdrop on the LTS and extract the CSI using special software (Picoscenes [43], Nexmon [44], etc.). We consider such a scenario where the victim, Bob, uses his mobile device (smartphone or tablet) to connect to Wi-Fi access points (APs) with shared passwords or no password protection. This is a reasonable assumption in public places such as libraries, office buildings, airports, and malls, where Wi-Fi access is often provided for the convenience of customers or visitors. AP (Alice) can prevent eavesdroppers (Eves, see Fig. 1) from stealing Bob's location while maintaining the legitimate user Bob's communication and localization capabilities. As an attacker, Eve is merely an ordinary user with a Wi-Fi network card.

In our threat model, Eve infers private information (such as current location information, trajectory, physical signs, and gestures) of nearby victim Bob by eavesdropping on CSI vectors. To focus on defense models, WiShield considers gesture recognition, respiratory monitoring, and trajectory tracking tasks. Moreover, Bob and Eve used publicly known gesture recognition model (Widar3.0 [45]), respiration monitoring model (MUSE-Fi [25]), and indoor localization models (e.g., mD-Track [3], Widar 2.0 [5] and SpotFi [4]) to perform human activity tracking. Finally, AP is always considered a trusted party. Specifically, our threat model gives attacker Eve the following capabilities:

Attacker position: We assume that a hostile eavesdropper, Eve, is interested in monitoring the activities of victim Bob in a closed space (home, conference room, etc.). Eve can initiate eavesdropping anywhere in the environment that can connect to an AP, so we do not impose restrictions on Eve's location. We allow Eve to choose any suitable eavesdropping location to maximize their chances of eavesdropping.

Attacker hardware: To achieve its goal, eavesdropper Eve uses a regular Wi-Fi network card (for example, the one used in laptops). We assume that Eve is configured with two antennas to eavesdrop on the CSI received by Bob's device to infer Bob's activity. It should be emphasized that Eve eavesdrops on the CSI information transmitted between Bob's device and the AP, so using more antennas will not enhance Eve's attack capabilities.

Attacker algorithm: To follow the principle of open design, which states that the security of a mechanism should not depend on the secrecy of its design or implementation. We let Bob and Eve use the same indoor localization models. In our evaluation, one gesture recognition model, Widar3.0 [45], one respiration monitoring model, MUSE-Fi [25], and three commonly used indoor localization models, mD-Track [3], Widar 2.0 [5] and SpotFi [4], were used to quantify the effectiveness of WiShield against malicious eavesdropping.

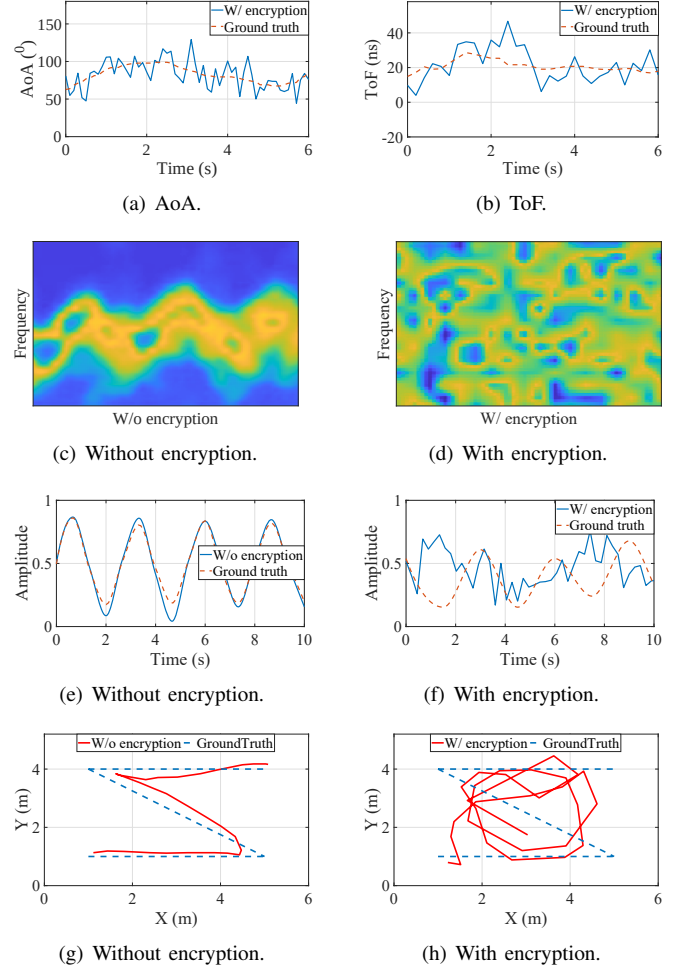


Fig. 3: Examples of CSI sensing results with or without encryption. (a)-(b) AoA and ToF are disrupted after encryption, (c)-(d) gesture recognition action in the frequency domain, (e)-(f) respiration waveform in the time domain, and (g)-(h) human trajectory tracking. After encryption, the CSI sensing results will be confused.

C. Feasibility Study

We use simple experiments here to further introduce the design of WiShield and verify its feasibility.

CSI obfuscation: As discussed in Section III-B, the eavesdropper, Eve, cracks the AP to obtain the CSI vector $\mathbf{h} = [h_m]$ and then executes the attacker algorithm to infer the sensitive activities of the victim, Bob. The basic idea of WiShield is to use MIMO beamformers [46] to encrypt the CSI vector $\mathbf{h}$ to prevent Eve from performing sensitive activity inference.

Specifically, we use an encryption matrix Ψ to scramble the LTS vector $\mathbf{l} = [l_m]$, the outcome $\mathbf{l}' = \mathfrak{d}(\mathbf{l}, \Psi)$ caused the CSI vector $[h_m]$ encryption as:

$$h_m = y_m / l'_m, \quad (2)$$

After the above encryption process, Eve cannot use the eavesdropped CSI to perform sensitive activity inference without the key of the encryption matrix Ψ . We use a simple case to briefly describe this encryption process, taking an LTS vector $\mathbf{l}$ of 4 transmitting antennas as an example. The component of each instance in the encryption matrix Ψ is 4 complex values corresponding to 4 antennas at the transmitting end, and $\mathbf{l}$ is multiplied to obtain 4 independently scrambled transmissions through the 4 transmitting antennas of the AP, which are received by the Rx antennas of Bobs and Eve.

We set up a few simple experiments to study the impact of CSI confusion on sensing applications. We used two 4-antenna USRPs as APs and another 4-antenna USRP as Eve. We first show the results of CSI confusion for parameter estimation, as shown in Fig. 3(a) and Fig. 3(b). After using CSI confusion, AoA and ToF are disrupted. We then demonstrate the impact of signal confusion on the gesture recognition task. We show the results of a push and pull action in the frequency domain. After signal encryption, the results in Fig. 3(c) are completely confused to the point of being indistinguishable as shown in Fig. 3(d). Next, we show the impact of signal confusion on respiratory monitoring. The monitoring results in Fig. 3(e) are very close to the ground truth, but in Fig. 3(f), effective respiratory waveforms cannot be obtained after signal confusion. Finally, we use the mD-track algorithm as the trajectory tracking algorithm to compare the trajectories after CSI obfuscation. Fig. 3(g) shows the trajectory recovered without using the encryption matrix Ψ . We further use the encryption matrix Ψ to perform CSI obfuscation to obtain Fig. 3(h). Obviously, after applying Ψ , the trajectory is disrupted, and Eve cannot recover the trajectory without the key.

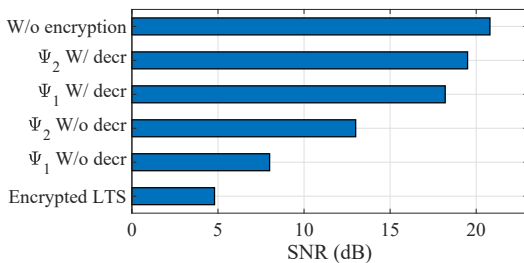


Fig. 4: Using channel encryption will reduce the SNR of the signal received at the Rx end, and using different decryption matrices can compensate for channel loss to varying degrees.

Impact on legitimate users: Although using the encryption matrix Ψ can effectively resist illegal eavesdropping, this method may reduce the communication quality of legitimate users. We use a set of experiments to demonstrate the impact of encryption and decryption on communication quality. We fixed the transmission power of the RF front-end to 20 dBm and the gain of the 4-antenna USRP end to 3 dBi. We use the AP to transmit 1000 data packets with encrypted LTS,

and another antenna node observes the signal-to-noise ratio (SNR) distribution of the LTS signal. As shown in Fig. 4, the average SNR of the encrypted LTS signal is 4.8 dB, which is lower than the 5 dB usually required by Wi-Fi networks. When the signal-to-noise ratio is lower than 5 dB, the error rate of LTS increases [47] significantly. Meanwhile, We pass the key Ψ to the legitimate user to decrypt the CSI. It is worth noting that using Ψ cannot completely restore the CSI because the encryption process may cause irreversible damage to the CSI. of distortion. We used two different Ψ to conduct experiments to observe the impact of decryption on SNR. From the results shown in Fig. 4, decryption can effectively improve the average SNR, but it is still lower than normal communication. In addition, the average signal-to-noise ratio of Ψ_2 is higher than Ψ_1 , which shows that we can improve SNR by selecting an optimal Ψ .

IV. THE SYSTEM DESIGN OF WISHIELD

A. System Overview

In this section, we briefly outline the overall workflow of WiShield. As the signal sender, the AP controls the entire indoor Wi-Fi link and uses the encryption matrix Ψ to encrypt the channel information. The AP sends the key to legitimate sensing users through the key distribution policy, and any illegal eavesdropper without the key cannot use the encrypted channel information to conduct privacy attacks. In Section IV-B, we detail our encryption process and propose a new distortion metric method. Then we introduce our encryption interval settings in Section IV-C. In Section IV-D, we propose a multi-objective optimization strategy to make the system achieve a balance between encryption strength, sensing accuracy, and communication quality. Finally, in Section IV-E, we introduce our CSI encryption and decryption process and key distribution strategy in detail, and in Section IV-F, we use the MLE algorithm to estimate parameters from the decrypted CSI.

B. CSI Time Series Encryption

The core of WiShield is to use matrix Ψ to encrypt Wi-Fi CSI to prevent the attacker Eve from inferring Bob's private information through Wi-Fi sensing. In this section, we will introduce our encryption strategy in detail. In order to better characterize the encryption strategy, we only consider one Wi-Fi NIC when introducing the encryption strategy (this strategy can be easily extended to cover multiple NICs). Specifically, we consider a CSI matrix consisting of the $M \times M$ beamformer matrix of the Q -th Tx antenna:

$$\mathbf{H} = [\mathbf{h}_1, \dots, \mathbf{h}_q, \dots, \mathbf{h}_{Q^{Tx}}], \quad (3)$$

The corresponding encryption matrix can be expressed as:

$$\Psi = [\Psi_1, \dots, \Psi_q, \dots, \Psi_{Q^{Tx}}]^T, \quad (4)$$

where $\Psi_q = \text{diag}(d_m)_q$ denote the beamformer matrix of size $M \times M$ applied to the q -th transmitting antenna, and d_m represent a distinct complex value pertaining to the m -th

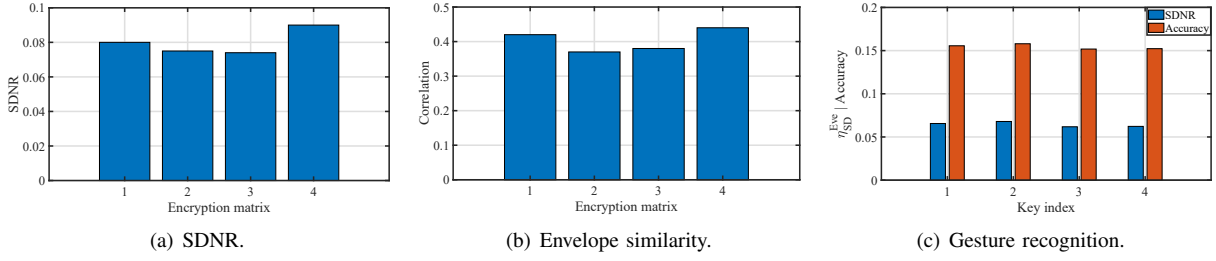


Fig. 5: (a)-(b) are correlation between four different encrypted SDNRs and corresponding signals, (c) is gesture recognition accuracy under four different Ψ

packet (CSI). Based on Eq. (3) and Eq. (4), the distorted CSI obtained at the receiving antenna can be expressed as:

$$\hat{\mathbf{H}} = \mathbf{H}\Psi, \quad (5)$$

It should be noted that we encrypted M CSI time series, which means the key length is M . To measure the effectiveness of the encryption process, we utilize the signal-to-noise ratio (SNR) to quantify relative entropy. Although SNR is a well-known signal quality metric in communications, we need to improve it to suit sensing tasks. Based on Eq. (1), the dynamic component h_m^D related to human movement can be expressed as:

$$h_m^D = \sum_{k \in K_D} \alpha_{m,k}^D e^{-j2\pi f_c \tau_k^D}, \quad (6)$$

and the static component h_m^S of environmental reflection in the wireless channel is:

$$h_m^S = \sum_{k \in K_S} \alpha_{m,k}^S e^{-j2\pi f_c \tau_k^S}, \quad (7)$$

Our modified SNR is defined as follows:

$$\eta_S = \frac{1}{M} \sum_{m=1}^M \frac{|h_m^D|^2}{|h_m^S|^2 + \sigma^2}, \quad (8)$$

where σ^2 represents the variance of Gaussian white noise. From the Eq. (8), we can observe that when the value of η_S is larger, the dynamic component accounts for a larger proportion, which means that the dynamic reflection signal is stronger and the sensing effect is better. However, WiShield's encryption process is based on MIMO, and Eq. (8) does not consider the distortion caused by MIMO encryption. We use here a new signal distortion measurement scheme called signal-to-distortion-plus-noise ratio (SDNR), we use $\hat{h}_m$ to present encrypted CSI and the SDNR η_{SD} can be expressed as:

$$\eta_{SD} = \frac{1}{M} \sum_{m=1}^M \frac{|\hat{h}_m^D|^2}{|\hat{h}_m - h_m|^2 + |h_m^S|^2 + \sigma^2}, \quad (9)$$

where $|\hat{h}_m - h_m|^2$ represents the distortion caused by MIMO encryption and also represents the entropy gain. To verify the effectiveness of SDNR, we use a simple experiment to calculate η_{SD} of the original and encrypted CSI. As shown in the Fig. 5(a) and (b), η_{SD} and The similarity of CSI signals

has a similar trend. To verify the relationship between η_{SD} and sensing quality. Taking the gesture recognition task as an example, Fig. 5(c) shows the average η_{SD} and gesture recognition accuracy under four different Ψ , which shows the strong correlation between η_{SD} and gesture recognition accuracy. These results prove that SDNR can effectively measure Wi-Fi sensing quality.

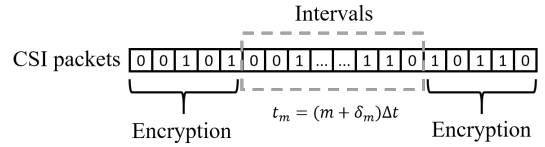


Fig. 6: WiShield uses randomized encryption intervals to increase encryption strength.

In our attack model, h_m is mainly used by the attacker Eve to infer Bob's private information, so the perceived information quality stolen by Eve due to MIMO encryption can be expressed as:

$$\eta_{SD}^{\text{Eve}} = \frac{1}{Q^{Tx}M} \frac{\|\hat{\mathbf{H}}^D\|_2^2}{\|\mathbf{H} - \hat{\mathbf{H}}\|_2^2 + \|\mathbf{H}^S\|_2^2 + \sigma^2}, \quad (10)$$

where process $\|\cdot\|_2$ represent l_2 norm of all Q^{Tx} CSI sequence. Since Eve did not get any information about the key Ψ (the amount is $Q^{Tx} \times M$). To obtain more effective observations for Bob's private information inference, Eve will try as many antenna combinations as possible to reduce the impact of the key Ψ . However, based on Eq. (3) and Eq. (5), WiShield will encrypt all CSI channels, which means that each new antenna will introduce new encryption. As a result, Eve cannot obtain accurate CSI by observing more antenna observations without explicit key Ψ information.

C. Encryption Time Confirmation

In order to reduce channel loss and increase encryption strength, WiShield uses random encryption time for encryption. Considering that encryption models are usually encrypted at a known and constant input interval, to use randomized encryption time, we use the encryption time strategy as shown in the Fig. 6. Specifically, to perform randomized encryption time, we control the interval between two encryption times as:

$$t_m = (m + \delta_m) \triangle t, \quad (11)$$

where $\delta_m \in [-1, 1]$ represent the randomization ratio for the m -th packet and $\triangle t$ is a constant period of time. Using this irregular encryption time strategy, Eve cannot perform CSI sequence recovery without a pre-trained neural network.

D. System Optimization



Fig. 7: The key distribution strategy WiShield used, Eve got obfuscated CSI.

Although the use of MIMO encryption can effectively defend against sensing attacks, as mentioned in Section III-C, even using key decryption may cause irreversible loss of channel information. As shown in Fig. 7, the AP distributes the encryption matrix Ψ to Bob through the common key distribution mechanism [48], but no matter which Ψ is used, the impact on the communication quality and perception tasks will exist. In order to balance the system's security and communication quality, we propose a multi-objective optimization framework here to defend against Eve's illegal eavesdropping on the premise of protecting the communication quality and perception accuracy of legitimate user Bob.

For a standard Wi-Fi system, APs obtain their CSI through Bob's standard feedback [49], and WiShield needs to obfuscate the CSI while delivering the key to Bob. Therefore, the optimization framework needs to integrate multiple objectives to achieve it. In this optimization framework, there are mainly legitimate sensing user Bob^S, legitimate communication user Bob^C, and attacker Eve. We use the encryption matrix Ψ as the variable to construct a multi-objective optimization model as:

$$\max_{\Psi} [-\eta_{SD}^{\text{Eve}}, \quad \eta_C^{\text{Bob}}, \quad \eta_{SD}^{\text{Bob}}], \quad (12)$$

where η_C^{Bob} represent the communication SNR of legitimate communication user Bob^C. $-\eta_{SD}^{\text{Eve}}$ and η_{SD}^{Bob} represent the SDNR of the attacker Eve and the legitimate sensing user Bob^S respectively. To achieve the optimization, Eq. (12) also needs to satisfy two constraints. First, the total power needs to satisfy:

$$\text{s.t. } \|\Psi\|_2^2 \leq MQ^{\text{Tx}}, \quad (13)$$

Second, the performance of Bob^C and Bob^S need to satisfy:

$$\eta_C^{\text{Bob}} \geq \epsilon_C, \quad \eta_{SD}^{\text{Bob}} \geq \epsilon_{SD}, \quad (14)$$

where ϵ_C and ϵ_{SD} represent the lower-bounds. According to the definition of SNR, η_C^{Bob} can be expressed as:

$$\eta_C^{\text{Bob}} = \|\hat{H}\|_2^2 / \sigma^2, \quad (15)$$

Based on Eq. (10), We can express η_{SD}^{Bob} slightly modified as:

$$\eta_{SD}^{\text{Bob}} = \frac{1}{Q^{\text{Tx}}M} \frac{\|\tilde{H}^D\|_2^2}{\|\Psi_q^{-1}\sigma\|_2^2 + \|\tilde{H}^S\|_2^2 + \sigma^2}, \quad (16)$$

where Ψ^{-1} is the reciprocal of Ψ , $\tilde{H}^S$ and $\tilde{H}^D$ are the decrypted static and dynamic channel reflection information related to Bob respectively. It is worth noting that $\|\Psi_q^{-1}\sigma\|_2^2$ represents the residual distortion after decryption. We use a simple experiment to demonstrate how different encryption matrices Ψ cause signal distortion. As shown in the Fig. 8, Fig. 8(a) is the original CSI signal, and Fig. 8(b)-8(d) represent the signals decrypted by different encryption matrices. Compared with the original signal, different Ψ will cause different degrees of signal distortion. Therefore, we need to optimize the encryption matrix Ψ . In order to find the best trade-off point between encryption strength, sensing accuracy and communication quality. We use a set of weights to express this set of relationships:

$$\max_{\Psi} \omega_1 \eta_C^{\text{Bob}} + \omega_2 \eta_{SD}^{\text{Bob}} - \omega_3 \eta_{SD}^{\text{Eve}}, \quad (17)$$

where $\sum_i \omega_i = 1$ and $\omega_i \in [0, 1]$. In order to obtain the optimal solution of Eq. (17), we use ADMM [50] to obtain the local optimal solution of this nonlinear non-convex optimization. In order to verify the effectiveness of the multi-objective optimization framework we designed, we randomly generated 200 Ψ and used them as initial solutions. We use the correlation of CSI before and after restoration as a measurement index. As shown in Fig. 9(a), after optimization, the CSI obtained by Eve has stronger encryption. On the contrary, Fig. 9(b) shows that after optimization, Bob^S obtains a signal closer to the original CSI. Finally, we use the bit error rate (BER) to represent the communication quality. The results in Fig. 9(c) show that the optimization strategy brings lower BER compared to random encryption. These results show that the multi-objective optimization strategy used by WiShield can achieve a good balance between encryption strength, sensing accuracy and communication quality.

E. CSI Encryption and Recovery

In order to preserve the sensing capabilities of legitimate users when defending against privacy attacks by illegal attackers, we design a dynamic neural network for CSI decryption. However, we still face two problems when designing the network. First, there may be multiple legitimate sensing users, but the encryption method of each user may be random, so each legitimate user may correspond to a specific decryption key. Secondly, the decrypted CSI may be distorted, which may cause catastrophic accuracy reduction for fine-grained sensing tasks and positioning tasks. Therefore, we need to optimize the decrypted CSI to ensure that it can perform sense task with high accuracy.

The neural network framework used by WiShield mainly consists of two main components: i) a model that can decrypt

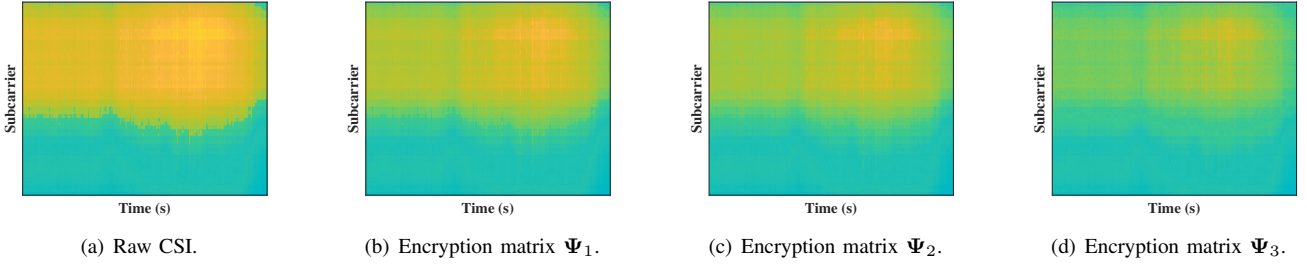


Fig. 8: Compared with the original CSI (a), the CSI signals restored using different encryption matrices (b)-(d) have varying degrees of distortion.

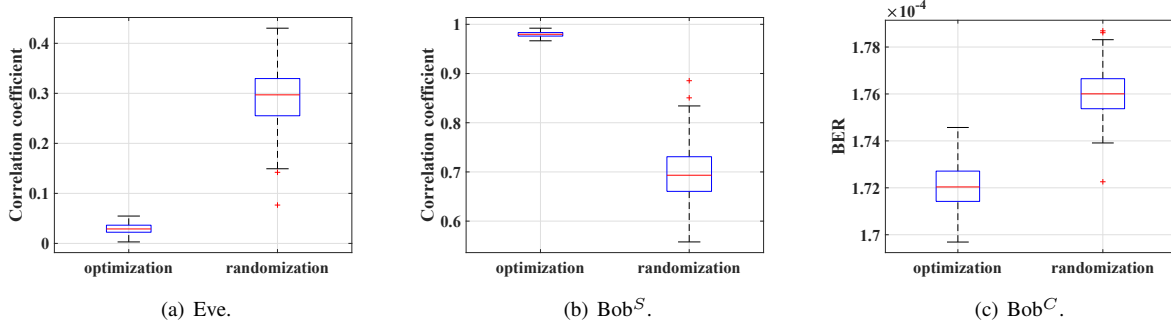


Fig. 9: Performance under different sensing subjects (Eve, Bob^S, and Bob^C) using optimized and randomized encryption strategies.

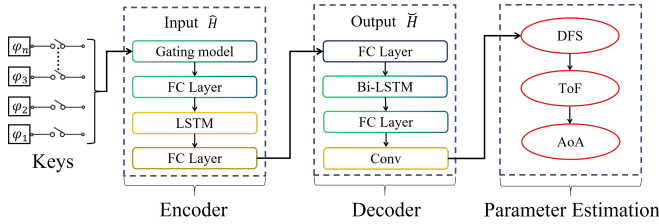


Fig. 10: Neural network structure of WiShield.

encrypted CSI, and ii) a neural network for recovering fine-grained CSI. We first introduce the CSI decryption model, as shown in Fig. 10, we use $\hat{H}$ to represent the restored CSI as the output of the model. The input to the model is the encrypted CSI $\hat{H}$ and the key φ . In the model, the AP securely shares the key φ with Bobs through a normal cryptographic hash function [51]. It is worth noting that there may be multiple legitimate users in the model, so the model needs to be able to adaptively distribute the corresponding keys φ to the encrypted CSI generated by different Ψ . To solve this problem, we borrowed the keyed dynamic neural network (KDNN) [52], KDNN can adaptively adjust the network structure and parameters according to different conditions of input data through a dynamic keying mechanism, making the network more flexible. This flexibility allows KDNN to perform well when dealing with various types of data and tasks. As shown in the Fig. 10, we use the set $[\varphi_1, \varphi_1, \dots]$ to provide additional information to control the gate. Using parallel ideas, each key φ_n as a separate sub-module, and switched to the corresponding version when processing the encrypted CSI input. The specific neural network structure is

shown in the Fig. 10, which includes an encoder and a decoder.

Encoder: The purpose of the encoder is to decrypt and recover the primary matrix from the encrypted CSI $\hat{H}$. First, the encrypted CSI $\hat{H}$ and φ are concatenated and sent to Fully connected layer. We then use a two-layer long short-term memory (LSTM) network [53] to generate continuous one-dimensional sequences. The dropout probability of LSTM is set to 0.5, and the hidden size is set to 256. Finally, we input the output of LSTM into a fully connected layer and input the result into the decoder.

Decoder: The purpose of the decoder is to recover the original CSI waveform from the output of the encoder. The encoder consists of a fully connected layer, followed by a bidirectional LSTM [54] with hidden sizes of 256 and 5, and an dropout probability of 0.5. The final output is sent to a fully connected layer and a convolutional layer to output the restored CSI $\hat{H}$. In Fig. 11, we show the feature maps of different CSI inputs corresponding to a gesture action. It can be found that the encrypted CSI obtains very different feature maps from the original CSI, but the restored CSI is almost close to the original CSI.

F. Parameter Estimation.

We formulate the CSI multi-parameter estimation problem as a maximum likelihood estimation (MLE) [55] problem. We use $\theta_l = (\alpha_l, \tau_l, \phi_l, f_{Dl})$ to represent the signal parameters of the l -th path. Where α_l , τ_l , ϕ_l and f_{Dl} represent attenuation, ToF, AoA and DFS respectively. Given the decrypted CSI vector h_m , we represent $\Theta = (\theta_l)_{l=1}^L$ as the MLE of multi-dimensional multi-path signal parameters. The log-likelihood function of Θ can be expressed as:

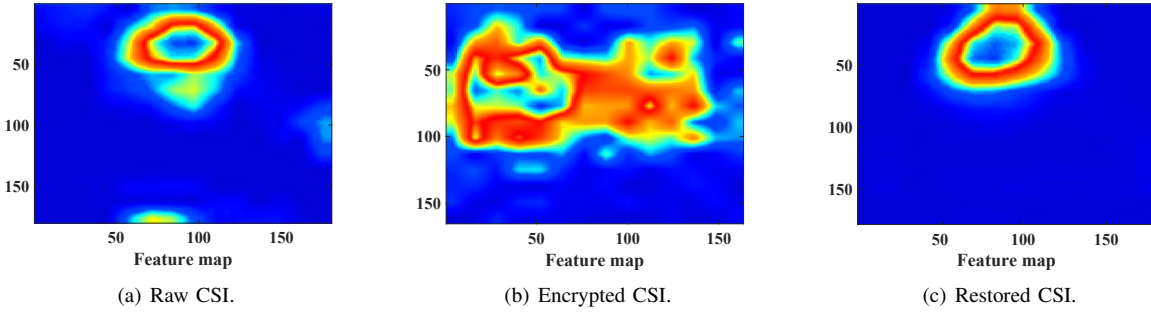


Fig. 11: Encryption and decryption are compared with the original CSI feature maps. The encrypted feature map is very different from the original feature map, but the decrypted feature map is close to the original feature map.

$$\Lambda(\Theta; h) = - \sum_{\mathbf{m}} \left| h(\mathbf{m}) - \sum_{l=1}^L P_l(\mathbf{m}; \theta_l) \right|^2, \quad (18)$$

where P_l is the signal if the l -th path, then the MLE maximization solution of Θ is expressed as:

$$\hat{\Theta}_{\text{ML}} = \underset{\Theta}{\text{argmax}} \{ \Lambda(\Theta; h) \}, \quad (19)$$

Unfortunately, this function is nonlinear, meaning there is no closed-form solution. In addition, the dimension of Θ (i.e., $4L$) is high, and it is very difficult to directly use Θ_{ML} to search when L is large.

In order to obtain the solution of the function, we need to reduce the overall search space, so we apply the spatial alternating generalized expectation maximization (SAGE) [56] algorithm. Each iteration of the algorithm only needs to re-estimate a subset of Θ components, and the estimation results of other components can be kept unchanged. So we can split the estimate of Θ into multiple estimates of a single parameter. Specifically, we need to optimize the parameters of each path in turn. For the l -th path, we first decompose the CSI, and then calculate the signal P_l of the l -th path:

$$\hat{P}_l(\mathbf{m}; \hat{\Theta}') = P_l(\mathbf{m}; \hat{\theta}_l') + \beta_l \left(h_{\mathbf{m}} - \sum_{l'=1}^L P_l(\mathbf{m}; \hat{\theta}_{l'}') \right), \quad (20)$$

Where $\hat{\Theta}'$ is the parameter obtained after the last iteration, β_l is the coefficient that controls the convergence speed of the algorithm and is set to 1 by default. The specific parameter estimation steps are as follows:

$$\begin{aligned} \hat{\tau}_l'' &= \underset{\tau}{\text{argmax}} \left\{ \left| z(\tau, \hat{\phi}_l', \hat{f}_{D_l}'; \hat{P}_l(\mathbf{m}; \hat{\Theta}')) \right| \right\} \\ \hat{\phi}_l'' &= \underset{\phi}{\text{argmax}} \left\{ \left| z(\hat{\tau}_l'', \phi, \hat{f}_{D_l}'; \hat{P}_l(\mathbf{m}; \hat{\Theta}')) \right| \right\} \\ \hat{f}_{D_l}'' &= \underset{f_D}{\text{argmax}} \left\{ \left| z(\hat{\tau}_l'', \hat{\phi}_l'', f_D; \hat{P}_l(\mathbf{m}; \hat{\Theta}')) \right| \right\} \\ \hat{\alpha}_l'' &= \frac{z(\hat{\tau}_l'', \hat{\phi}_l'', \hat{f}_{D_l}'', \hat{P}_l(\mathbf{m}; \hat{\Theta}'))}{TFA}, \end{aligned} \quad (21)$$

where

$$z(\tau, \phi, f_D; P_l) = \sum_{\mathbf{m}} e^{2\pi \Delta f_j \tau_i} e^{2\pi f_c \Delta s_k \cdot \phi_i} e^{-2\pi f_{D_l} \Delta t_i} P_l(\mathbf{m}), \quad (22)$$

where the MLE of α_l can be derived in closed form as a function of τ_l , ϕ_l and f_{D_l} , so is calculated at the end of each iteration.

V. IMPLEMENTATION AND SETUP

In this section, we introduce the hardware implementation of WiShield, followed by the experimental setup and metrics.

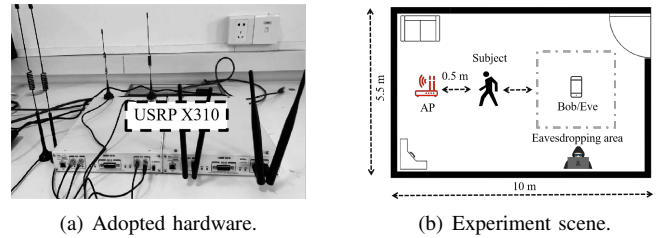


Fig. 12: (a) Hardware configurations and (b) experiment scene of WiShield.

System Implementation. As shown in Fig. 12(a), we use the widely adopted USRP X310 [57] for Wi-Fi sensing to build our WiShield prototype. USRP X310 works at 6GHz, 200MHz bandwidth, and transmits data packets at a default rate of 1000 data packets. Bob and Eve are configured with 4 antennas, respectively, and the AP is configured with 4 antennas. All USRP nodes are connected via Ethernet switches to PC servers powered by Intel Core i9-12900K CPUs, 32GB of memory, and RTX 2080 graphics cards. We implemented the software framework of WiShield and Python 3.7 on the PC server, and the neural model was built on PyTorch 1.11.0.

Experiment Setup. We recruited 20 subjects (13 females and 7 males) to participate in data collection. All subjects were asked to watch a sample video to standardize each volunteer's gesture and trajectory demonstration behavior. We perform 10 common gestures (i.e. push and pull, slide, clap, draw circle, square, zigzag, Ok, Pinch, Cross and Swap) in a large conference of 55 square meters. For ease of explanation, we abbreviate these gestures as G1, G2, G3, G4, G5, G6, G7, G8, G9 and G10. We collected encrypted and unencrypted data for each volunteer, where the encrypted data is 16000 samples (10 gestures $\times$ 80 instances $\times$ 20 subjects), and the unencrypted data is 10000 samples (10 gestures $\times$ 50 instances $\times$ 20 subjects). Then we collected respiratory data for up to

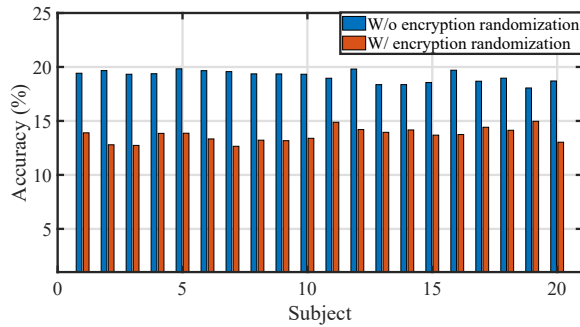


Fig. 13: The random encryption time strategy reduces the accuracy of gesture recognition tasks.

20 minutes from each subject using the same principle, and the respiratory rate was divided into three rates. For trajectory tracking, each subject was asked to follow the instructions to record data for two trajectories (circular and Z-shaped) and four random lines, and each trajectory was performed 20 times. Our data collection process strictly adheres to our institution's IRB requirements.

Metrics. WiShield uses two commonly used metrics to quantify sensing and communication performance, namely accuracy and Bit Error Rate (BER). Where accuracy represents the proportion of correctly recognized gestures, and the bit error rate represents the proportion of incorrectly decoded bits. For respiration monitoring, we use breath per minute estimation error (bpm) to evaluate the respiration detection error. Finally, we use localization error to evaluate the error of the trajectory tracking task.

VI. EVALUATION

In this section, we first evaluate the random encryption time strategy and model robustness. Then we comprehensively evaluate WiShield's defense capabilities against perceived attacks.

A. Benchmark Studies

1) *Encryption time randomization:* In Section IV-C, we propose a strategy that uses encryption time randomization to improve encryption strength. In order to verify the effectiveness of encryption time randomization, take the gesture recognition task as an example, we tested the accuracy of gesture recognition tasks on 20 subjects without using encryption time randomization (only using Ψ) and adding encryption time

randomization. The results depicted in Fig. 13 demonstrate that employing random time encryption optimization results in a further reduction in the accuracy of the gesture recognition task compared to not using it. This outcome suggests that the random encryption time strategy enhances the encryption strength of WiShield.

2) *Model robustness:* To verify the robustness of WiShield's neural network, we tested the neural network's ability to recover CSI under dynamic channel conditions. Specifically, we calculated the similarity between the encrypted and decrypted (LoS and NLoS) CSI and the original CSI respectively. As shown in Fig. 14, the average correlation after decryption is above 98%, which also explains that the CSI after the system can accurately perform sensing applications.

B. WiShield Performance

1) *Defense capability:* We take three common Wi-Fi sensing applications as examples to evaluate WiShield's defense capabilities against malicious eavesdropping. We first conducted a gesture recognition experiment, where we asked 20 subjects to make different gestures at 0.5 m from the AP and first fixed Eve at a position 2 m from the AP. We set up two sets of experiments, one with no obstacles between Eve and AP and the other with NLoS-type added obstacles. As shown in Fig. 15(a), after encryption, the average gesture eavesdropping accuracy under LoS conditions is less than 16%. Compared with unencrypted eavesdropping, WiShield successfully reduced the eavesdropping accuracy by more than 80%. In addition, experiments under NLoS conditions also have a similar trend, except that when obstacles are blocked, the gesture recognition accuracy will decrease slightly.

We then evaluate WiShield's defense capabilities against physical sign detection. We asked subjects to perform breathing experiments at a distance of 0.5 m from the AP. We also collected data under both LoS and NLoS conditions. As shown in the Fig. 15(b), after encryption, the error of breath detection is significantly increased to the point that it is no longer referenceable. These results show that WiShield can effectively prevent the eavesdropping of sensitive action information.

We finally evaluated WiShield's resistance to illegal location eavesdropping. In the experiment, subjects were asked to walk along a prescribed path. As shown in the Fig. 15(c), under the condition of no encryption, the positioning error is about 0.4 m. After encryption, the positioning error rises to more than 2.8 m. This positioning error is unacceptable indoors within 60 square meters. This result is consistent under both LoS and NLoS conditions, indicating that WiShield is effective in preventing illegal location eavesdropping.

2) *Different device's position:* We further studied the impact of the distance between Eve and AP on the results. We placed Eve at eight locations between 2 m-6 m away from the AP, with the step size set to 0.5 m. As shown in Fig. 16(a), the performance of gesture recognition decreases as the distance increases. Similar results are also shown in Fig. 16(b) and Fig. 16(c). The breathing error rate and positioning error will also increase as the distance increases. These results confirms that Eve's eavesdropping performance does deteriorate over longer distances.

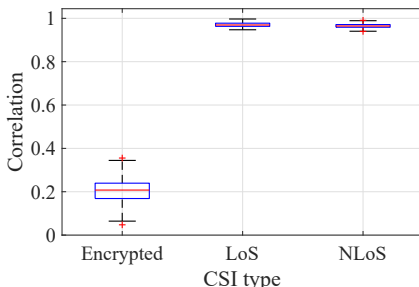


Fig. 14: CSI correlation.

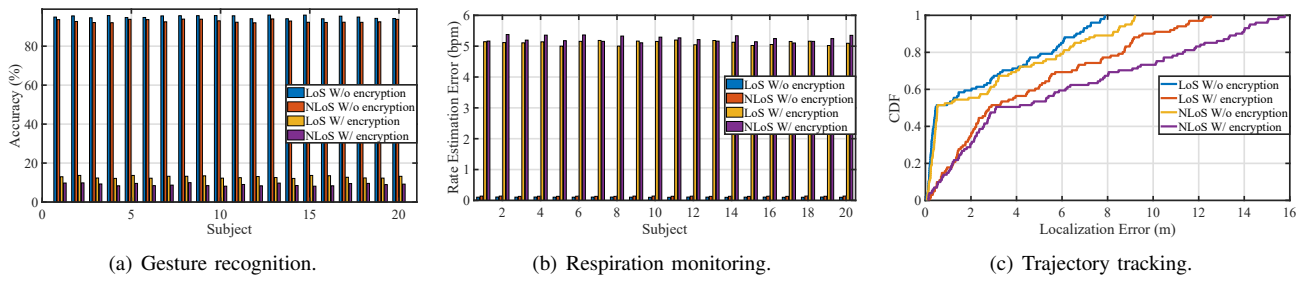


Fig. 15: Eve's eavesdropping performance evaluated of (a) gesture recognition, (b) respiration monitoring, and (c) trajectory tracking.

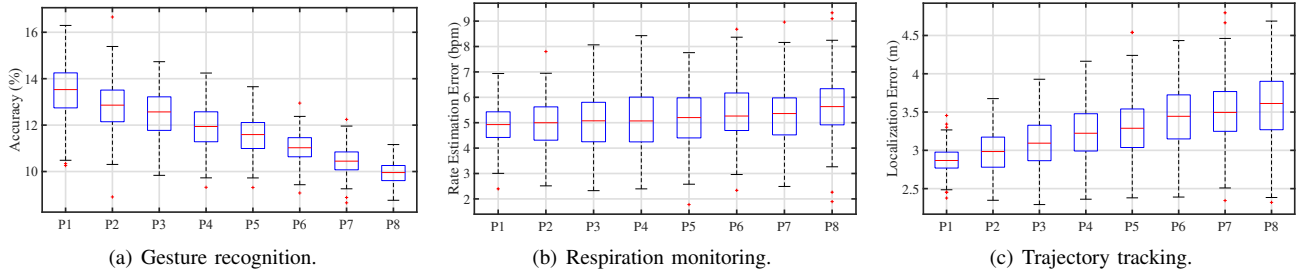


Fig. 16: The impact of different device's position.

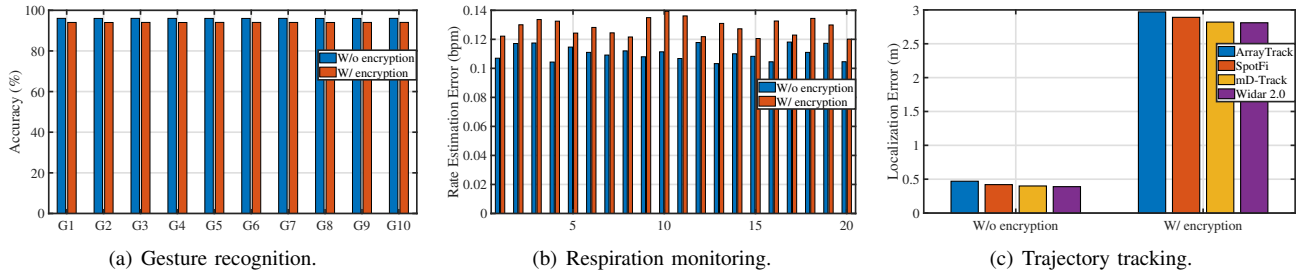


Fig. 17: Performance evaluation of Wi-Fi sensing applications after CSI decryption (a) gesture recognition, (b) respiration monitoring, and (c) trajectory tracking.

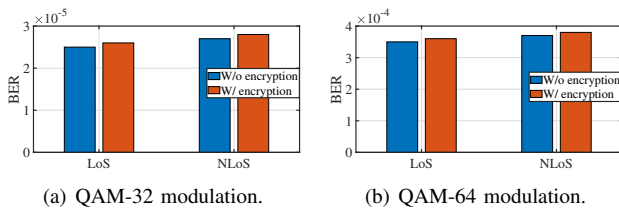


Fig. 18: The impact of the encryption process on the bit error rate under two different modulation schemes.

3) *Impact of encryption on sensing tasks:* In addition to effectively resisting sensing attacks, WiShield also needs to maintain the sensing capabilities of legitimate users. Therefore, we evaluate the performance recovery of sensing applications after CSI decryption under the same experimental setting. For comparison, we considered two settings: i) using unencrypted CSI for data collection and ii) using a model to encrypt and decrypt the CSI. Fig. 17(a) shows the average accuracy of gesture recognition before and after decryption.

For six common gestures, the average recognition accuracy before decryption is 92.5%, and the average recognition accuracy after decryption is 90.2%. This shows that the decryption process will indeed cause some loss of information, but it is still capable of gesture recognition tasks.

Next, we demonstrate the recovery accuracy of WiShield for human respiration monitoring. As shown in the Fig. 17(b), when it is not encrypted, the respiration rate estimation error is about 0.11 bpm. The error after decryption increases slightly, with an average of about 0.13 bpm. Overall, the distortion caused by decryption has little impact on breath detection.

Finally, we study the impact of encryption on trajectory tracking tasks. Unlike activity recognition tasks such as gesture recognition and physical sign detection, trajectory tracking requires more fine-grained CSI information, while activity recognition relies more on the classification ability of the model. We used four widely accepted trajectory tracking models, namely ArrayTrack, SpotFi, mD-Track and Widar 2.0, for experiments. The experimental results are as shown in the Fig. 17(c). When not encrypted, the positioning accuracy of the four methods is around 0.42 m to 0.49 m, and after

encryption, the positioning accuracy increases to 3.01 m to 2.79 m. This error occurs because for trajectory tracking tasks, the errors generated during CSI encryption will gradually accumulate as the information is distorted. At present, it seems that the encryption process does have an impact on positioning accuracy. We will continue to address this issue in future work.

4) *Impact to BER*: In this section, we study the impact of encryption on communication quality. We use the bit error rate of the payload as a metric. We take a communication time of 30 minutes as an example and use two common modulation methods (QAM-32 and QAM-64) as the payload. As shown in the Fig. 18, the encryption process has a very weak impact on communication performance. No matter which modulation method is used, the bit error rate is low enough. This is due to the multi-objective optimization framework we proposed, which successfully achieved encryption strength, sensing Balance of accuracy, and communication quality.

5) *Compare with related work*: We tested OpenWiFi CSI Fuzzer [58] and WiShield under the same conditions using the 4-antenna system as the benchmark. The results are as shown in Fig. 19. WiShield's encryption performance is better than OpenWiFi CSI Fuzzer's in gesture recognition, respiration monitoring, and trajectory tracking. Because our encryption applies an encryption key Ψ where each element is a complex number of $Q_{Tx} \times M$ flexibility, which takes advantage of control of the MIMO beamformer, allowing for greater strength and flexibility. Second, KDNN can capture the correlation between CSI flows between devices and generate keys in a more sophisticated way and can also be continuously optimized and improved to meet changing needs.

We also evaluate the performance of WiShield's key generation algorithms compared with similar past work. As shown in the Table. I, thanks to the use of KDNN, WiShield achieves a key generation rate of more than 70 bits/second in all evaluation scenarios. In addition, WiShield's key generation solution can be extended to security applications on similar IoT devices to combat covert radio frequency eavesdropping attacks.

6) *Antenna quantity*: Increasing the number of AP antennas can enhance the diversity of keys Ψ , thereby thwarting attackers' attempts to safeguard the interests of legitimate users. As shown in the Fig. 20, as the number of AP antennas increases, the performance of both ends of WiShield improves. However, when the number of antennas increases to 4, the improvement decreases significantly, so our experiment uses 4 antennas as the benchmark.

7) *Capability to thwart sensing attacks*: Wi-Fi-based gesture recognition can allow eavesdroppers to obtain sensitive information such as passwords, and we using WINK [31] and WindTalker [19] as baseline methods. Among them, WINK is a password inference strategy that does not require training, and WindTalker is a password inference and numeric key classification strategy that requires training. As shown in Fig. 21(a), WindTalker's single key classification accuracy dropped by 60% after encryption. Besides, Fig. 21(b) shows the password inference accuracy of WINK and WindTalker before and after encryption. It can be seen that encryption

TABLE I: The comparison between WiShield and prior work

	key-generation rate	real evaluation test
WiShield	70 bit/sec	yes
liu et al. [59]	40 bit/sec	yes
Wei et al. [60]	25 bit/sec	yes
Thai et al. [61]	8 bit/sec	no
Xu et al. [62]	6.2 bit/sec	no

greatly affects this type of privacy attack strategy based on Wi-Fi CSI.

8) *System overhead*: We prototyped WiShield using the widely adopted USRP X310, which operates at 6 GHz, 200MHz bandwidth, and transmits packets at a default rate of 1000 packets. Take the setting of AP's 4 antennas to send 2s (2000 packets) of traffic to legitimate sensing users as an example. The encryption matrix Ψ contains 4×2000 complex numbers, and each packet is about four bytes, so the total data volume is 32000 bytes. Fortunately, WiShield only requires a decryption key φ to activate a specific encoder to process encrypted CSI, and the transmission volume of the decryption key φ requires only a 2048-bit decryption key, which is much less than the original volume. And φ and Ψ only need to be updated regularly every few weeks. We tested the time required for each decryption operation in different user scenarios. When the number of users exceeds 4, the decryption time does not exceed 0.82 ms, and when the number of users is 8, the decryption time remains within 1.1 ms. On our prototype, each decryption operation related to Ψ takes 0.78 ms, so the additional time cost of the 2s data stream is $2000 \times 0.78 = 1560$ ms.

VII. DISCUSSION

In this section, we discuss the security, algorithm efficiency and application scenarios limitations of WiShield.

Security analysis. For attacker Eve, he has no idea about the complex-valued parameters involved in the key Ψ (the number is $Q_{Tx} \times M$), and he can only get more through different antennas Observations to crack Ψ . However, according to $\hat{H} = H\Psi$, every time Eve adds a new antenna, new parameters related to Q_{Tx} will be introduced and Ψ is unknown. Therefore, Eve cannot gain diversity benefits by using more antennas. Secondly, WiShield employs a scrambled pattern sequence and utilizes the sequence length as the key length for encryption, which stands in stark contrast to the traditional orthogonal blind scheme. In contrast to the latter, where the encryption key is constrained to maintain orthogonal channels with legitimate users to shield against attackers, WiShield allows the legitimate user to select the encryption key from the entire space. Moreover, WiShield's encryption keys typically consist of approximately or even more than 1,000 shuffled patterns, with rapid switching between patterns occurring on the order of milliseconds. As a result, WiShield's encryption method is significantly more robust than existing methods. Additionally, we also consider the possibility of leaking keys during periods when there is no sensing activity. According to Eq. (10), the static period can be considered as $\|H^S\|_2^2$ in the equation, which belongs to the static component. Since this

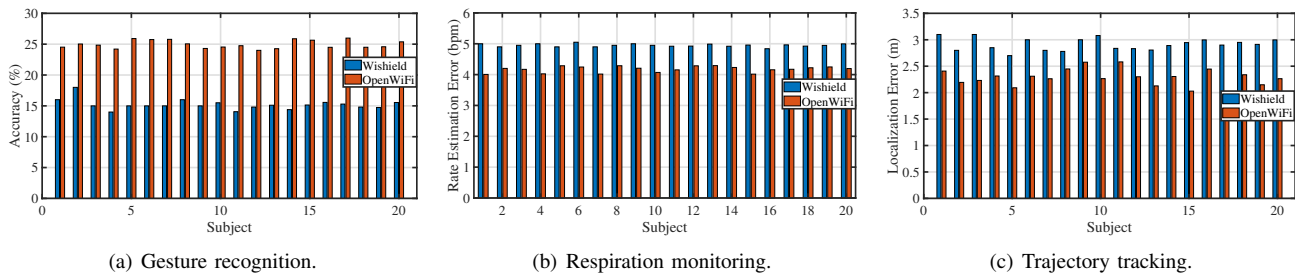


Fig. 19: Performance comparison of WiShield and OpenWiFi CSI Fuzzer.

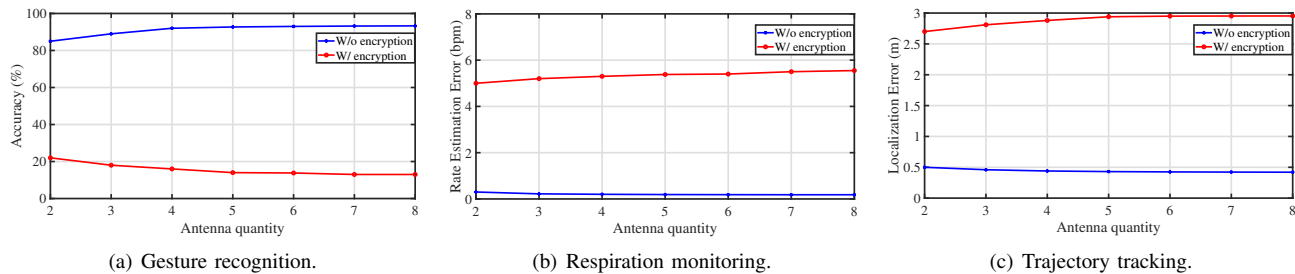


Fig. 20: Impact of antenna quantity on sensing applications, (a) gesture recognition, (b) respiration monitoring, and (c) trajectory tracking.

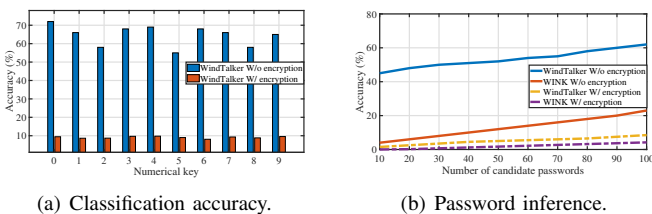


Fig. 21: Comparing classification accuracy and password inference on two classic keystroke inference strategies.

component is statistically irrelevant to sensing activity, it does not matter whether the CSI of this period is encrypted or not, so it will not affect the protection strength of WiShield during periods when there is no sensing activity.

Training overhead. WiShield only needs to perform the KDNN training process once for each registered legal user during the initialization phase of the system. Specifically, during the user registration phase, we train a unique KDNN for each legitimate user device based on the encrypted obfuscation function and the CSI data flow of the device and AP port. Then, we send the trained KDNN to the corresponding legal device through a secure channel. Since the obfuscation function is strictly confidential, WiShield ensures that eavesdroppers cannot infer the key through reconstruction attacks through the reconstructed CSI data stream. The training and key distribution of KDNN are only performed once during system initialization and do not need to be performed every time subsequent sensing applications are performed. Although WiShield uses a neural network, the computational load of this training is still lightweight for the calculations of today's commonly used IoT devices (mobile phones or tablets). In

addition, if necessary, we can offload the network initialization process to the server. Under this condition, whenever a sensing application is required, the user device only needs to perform a KDNN forward calculation and a simple quantification operation to obtain the key. Therefore, WiShield is a feasible privacy protection solution for commonly used indoor Wi-Fi devices.

Algorithm efficiency. While WiShield's encryption and protection can improve the security and reliability of Wi-Fi networks, implementing encryption and protection mechanisms often requires additional complexity and development work. This includes key management, authentication processes, encryption algorithm selection and implementation, etc. Complex implementations may make system design and maintenance more difficult, and for resource-constrained environments such as mobile devices, encryption and protection mechanisms may increase the device's power consumption. Especially when encryption algorithms are used for data encryption and decryption, more power resources may be consumed. To solve this problem, we set out to design a plug-and-play system rather than a one-size-fits-all approach to encryption. For application scenarios with lower security levels, we can adaptively adjust the encryption strength to reduce system power consumption.

Application scenarios. In AR/VR scenarios, WiShield needs to be activated to adapt to the scene requirements. For example, in gesture recognition, where security requirements are low, low latency is required, so this activity does not need to be encrypted. However, privacy is critical for activities requiring authentication or password entry, and WiShield can effectively protect user privacy. In applications that require precise localization, such as real-time interactions in virtual environments, encryption may not be suitable because there may be time delays and information distortion after decryption,

which may hinder feasibility. Similarly, for UAV navigation, if it is used for sensitive activities such as military defense, WiShield can protect sensitive trajectories well. Still, encryption may sacrifice certain efficiency for general civilian devices. In the Metaverse, the impact of WiShield varies depending on specific scenarios and applications. Users often expect precise interactions with virtual objects, such as object manipulation. Therefore, cryptographic sensing may be better for applications requiring high interaction accuracy. However, for highly private interactions, encryption is a must.

VIII. CONCLUSION

In this paper, we propose WiShield, a defense system specifically designed for ubiquitous Wi-Fi sensing attacks. WiShield innovatively uses a carefully designed key distribution strategy to achieve a balance between encryption strength, sensing accuracy, and communication quality through MIMO encryption. Unlike past methods, WiShield does not require the use of expensive additional equipment or complex system modifications. We have verified the effectiveness of WiShield through a large number of experiments, and the system we proposed can well protect legitimate sensing users. We believe that the proposal of WiShield has raised the standard of privacy protection in the field of Wi-Fi sensing.

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